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Ambiguity-selective consistency regularization for mean-teacher semi-supervised medical image segmentation

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ABSTRACT

Semi-supervised learning has greatly advanced medical image segmentation since it effectively alleviates the need of acquiring abundant annotations from experts, wherein the mean-teacher model, known as a milestone of perturbed consistency learning, commonly serves as a standard and simple baseline. Inherently, learning from consistency can be regarded as learning from stability under perturbations. Recent improvement leans toward more complex consistency learning frameworks, yet, little attention is paid to the consistency target selection. Considering that the ambiguous regions from unlabeled data contain more informative complementary clues, in this paper, we improve the mean-teacher model to a novel ambiguity-consensus mean-teacher (AC-MT) model. Particularly, we comprehensively introduce and benchmark a family of plug-and-play strategies for ambiguous target selection from the perspectives of entropy, model uncertainty and label noise self-identification, respectively. Then, the estimated ambiguity map is incorporated into the consistency loss to encourage consensus between the two models' predictions in these informative regions. In essence, our AC-MT aims to find out the most worthwhile voxel-wise targets from the unlabeled data, and the model especially learns from the perturbed stability of these informative regions. The proposed methods are extensively evaluated on left atrium segmentation and brain tumor segmentation. Encouragingly, our strategies bring substantial improvement over recent state-of-the-art methods. The ablation study further demonstrates our hypothesis and shows impressive results under various extreme annotation conditions.

1. Introduction

Medical image segmentation is a fundamental process in many image-guided therapies. Recently, deep learning (DL)-based methods have greatly advanced automatic organ or tumor segmentation (Li et al., 2018; Dou et al., 2016). However, their success usually relies on a large amount of labeled data, yet, the acquisition of which is generally laborious, expertise-demanding and even infeasible in institutes with limited labeling budgets. Considering that the unlabeled data is often abundant in practice, semi-supervised learning (SSL), which can utilize both labeled and unlabeled data, becomes an active research topic.

Considerable efforts have been made on semi-supervised segmentation, where most methods feature two popular fashions, i.e., pseudo labeling and consistency regularization. Firstly, being an intuitive way, **pseudo labeling** aims to generate pseudo labels for unlabeled images via the model primarily trained with limited labeled data and then mixes the new pseudo labeled data with the original labeled set for further optimization (Zhang et al., 2022; Zheng et al., 2020; Bai et al., 2017). Inherently, pseudo labeling-based methods attempt to mimic the typical supervised training process to leverage the unlabeled data. Since the labeled 3D medical images are scarce, the quality of pseudo labels cannot be guaranteed; thus, finding out reliable and correct voxel-wise pseudo labels is appreciated and critical in this training scheme (Rizve et al., 2021; Zhang et al., 2021a). In contrast, **consistency regularization**, which is based on the smoothness assumption (i.e., if x and x' are close in the input space, their labels y and y' should be the same (Van Engelen and Hoos, 2020)) explores an unsupervised manner to leverage the unlabeled data. Specifically, this fashion encourages the predictions to be consistent under different perturbations at data-level or model-level for the same input. For example, the Π model (Samuli

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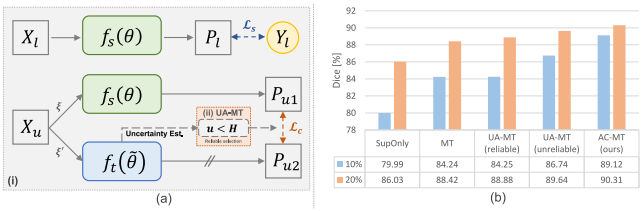

Fig. 1. (a) Illustration of the MT model (Tarvainen and Valpola, 2017; Cui et al., 2019) and the extended UA-MT model (Yu et al., 2019). $f_s(\theta)$ and $f_t(\tilde{\theta})$ are the student model and the EMA teacher model, '//' means stop-gradient, ξ and ξ' are perturbations, and P represents prediction. (b) Quantitative comparison on LA segmentation under the 10% and 20% labeled data settings. AC-MT (ours) indicates our best-performing model in this task (details in Section 4.2).

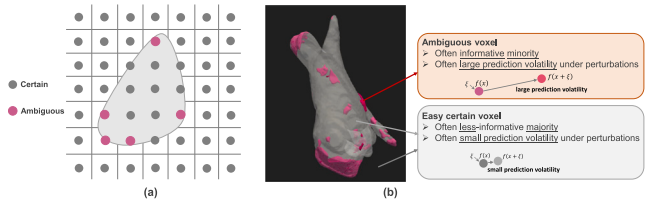

Fig. 2. (a) A 2D illustration of certain region and ambiguous region followed by (b) an example of the ambiguous map E (pink in 3D with fused GT) on LA segmentation estimated by the indicator of systematic label noise identification (see Section 3.3) under the 10% labeled setting. $f(\cdot)$ denotes the function; ξ indicates perturbations.

and Timo, 2017) encourages consistency of multiple forward predictions under different data-level perturbations. Further, the temporal ensembling strategy (Samuli and Timo, 2017) improves the Π model by utilizing *exponential moving average* (EMA) predictions for unlabeled data as the consistency targets. To alleviate the burden caused by maintaining the EMA predictions, an elegant design called the mean teacher (MT) model (Tarvainen and Valpola, 2017; Cui et al., 2019) introduces a self-ensembling teacher model efficiently updated by the EMA weights of the student model and leverages unlabeled data by encouraging consensus of the student's and teacher's outputs, taking both data-level and model-level perturbations into consideration, as shown in Fig. 1(a). Being a simple and efficient design, the MT architecture also becomes popular in other tasks, e.g., learning with mixed-quality labels (Wang et al., 2020a) and image registration (Bigalke et al., 2022; Xu et al., 2022b). Subsequent to the MT model, many consistency-based methods have been proposed with the majority focusing on more complex and often computationally demanding manners for extra supervision signals, e.g., adding several auxiliary branches to perform multi-task consistency (Luo et al., 2021; Wang et al., 2021), introducing contrastive learning for more self-supervision signals (You et al., 2022), transformation consistency (Li et al., 2020a) and cross-consistency training (Ouali et al., 2020).

Despite improvement brought by newly proposed complex models, unfortunately, little attention is paid to improving the vanilla MT model from the perspective of target selection, i.e., which voxels should be considered as the consistency targets. The most relevant work is UA-MT (Yu et al., 2019), as shown in the orange box in Fig. 1(a). UA-MT introduces Monte Carlo dropout (Kendall and Gal, 2017) to estimate model uncertainty from the teacher model and then only preserves the reliable (low-uncertainty) predictions when computing the consistency loss, i.e., only the vth voxel with $u_v < H$ in the student's and teacher's outputs are selected, where u and H are the uncertainty and an empirical threshold (e.g., from $0.75u_{max}$ to u_{max}) (Yu et al., 2019), respectively. However, only marginal improvement was achieved by UA-MT, as shown in Fig. 1 (b). Here, we revisit UA-MT and reverse its selection criterion (denoted as UA-MT (unreliable)) that encourages the consensus on the ambiguous voxels ($u_v > H$), the model even improves the original UA-MT (reliable) on the left atrium segmentation benchmark (Xiong et al., 2021; Yu et al., 2019) (Fig. 1(b)). This observation reveals a neglected problem that considering the "unsupervised" consistency learning in a "supervised" way may be suboptimal and even inappropriate. More reliable voxels are intuitively favored in the supervised-like fashion, e.g., pseudo labeling (Zhang et al., 2022; Zheng et al., 2020; Bai et al., 2017; Rizve et al., 2021), yet, learning from unsupervised consistency regularization is inherently learning from prediction stability under perturbations to encourage generalized representation and thus effectively alleviate overfitting on limited labeled data. This fashion does not require the pseudo "ground truth" for training and therefore the correctness of the predictions is not necessarily a concern. Thus, it is natural to ask: what does the model exactly want from the additional unlabeled data in unsupervised consistency regularization?

Intuitively, certain voxels often exist in less-informative regions with small prediction volatility under perturbations, which can already be well classified by the model (often the majority, as shown in Fig. 2) given the supervision from labeled data. These regions usually have small stability losses so that they potentially diminish the efficacy of ambiguous regions (often the minority) since the consistency loss commonly has voxel-wise average nature, e.g., mean squared error (MSE) (Tarvainen and Valpola, 2017). Instead, the hard-yet-informative regions (e.g., small branches or blurred edges around the left atrium, with an example in Fig. 2) often exhibit unreliable nature (Wu et al., 2021; Shi et al., 2021) with high prediction volatility under perturbations. Thus, it is natural to hypothesize that emphatically encouraging stability in these challenging ambiguous regions could be more rewarding for model training.

Based on the above observations, we further improve the classical MT architecture for semi-supervised medical image segmentation, where we put emphasis on boosting the performance via selecting productive unsupervised consistency targets. Specifically, we propose a simple-yet-effective ambiguity-consensus mean-teacher (AC-MT) model, as shown in Fig. 3, to better exploit the complementary informative clues from unlabeled data. Considering that uncertainty can characterize ambiguity (Czolbe et al., 2021), while it also correlates with error-prone regions, we comprehensively introduce a family of plug-and-play strategies for ambiguous (informative) target selection from the perspectives of entropy, model uncertainty and label noise self-identification, respectively, without introducing any trainable parameters or modifying the backbones. Then, rather than enforcing consistency on the easy certain regions (Yu et al., 2019; Hu et al., 2022; Adiga V et al., 2022), we believe that ambiguity consensus can bring more informative clues and thus the estimated ambiguity map is incorporated into the consistency loss to encourage consensus between the two models' predictions in these regions. Inherently, our method aims to find out and emphasize the most worthwhile (challenging) regions from unlabeled data. The model is only required to especially learn from the perturbed stability of these voxels, but never distracted by those already well-classified easy ones.

Overall, the main contributions of this work are as follows:

- We experimentally identify a neglected-yet-critical problem, i.e., consistency target selection, that limits the performance of classical mean-teacher semi-supervised segmentation model, and present a new perspective that encouraging perturbed stability in ambiguous-yet-informative regions can drive the model to learn more useful representations from unlabeled data.
- Based on the above perspective, we improve the mean-teacher model to a novel ambiguity-consensus mean-teacher (AC-MT) model, equipped with a comprehensive family of plug-and-play strategies for ambiguous (informative) target selection.
- Our methods are extensively evaluated on left atrium segmentation (Yu et al., 2019; Xiong et al., 2021) and brain tumor segmentation (Spyridon (Spyros) Bakas, 2020). Encouragingly,

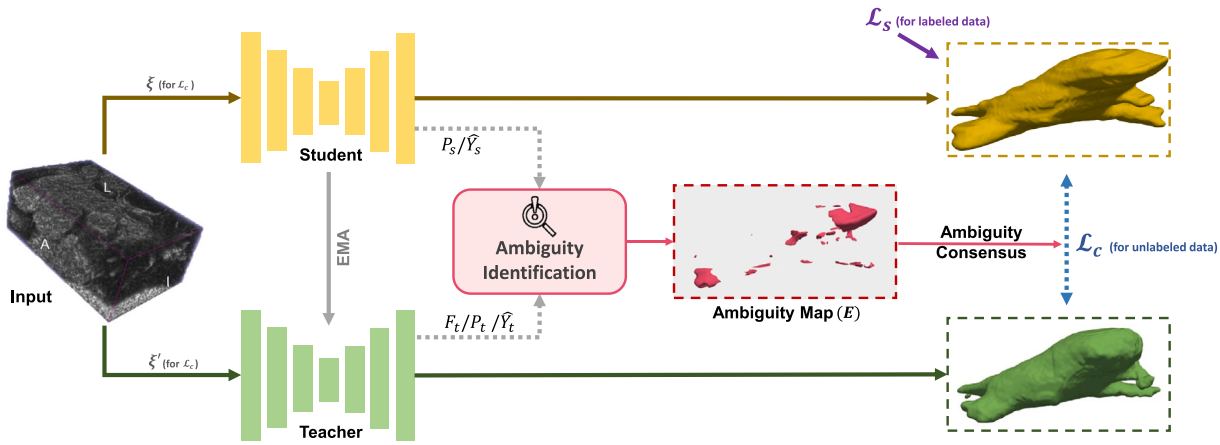

Fig. 3. Overview of our ambiguity-consensus mean-teacher (AC-MT) model, where P_s and $\hat{Y}_s$ are the predicted probabilities and the argmax pseudo labels from the student model, respectively; F_t, P_t and $\hat{Y}_t$ represent the feature maps from the layer before the penultimate convolution, the predicted probabilities from the self-ensembling teacher model and the argmax labels from the teacher model, respectively. The student model is updated by back-propagation. The teacher model is updated by exponential moving average (EMA) of the student's weights. E is the estimated ambiguity map. $\mathcal{L}_s$ is the supervised loss for labeled data and $\mathcal{L}_c$ is the ambiguity-masked consistency loss for unlabeled data.

such a simple strategy brings significant improvements over the vanilla MT and UA-MT (Fig. 1(b)), and also outperforms other lines of state-of-the-art semi-supervised methods, indicating that our approach can effectively find out the most worthwhile targets to learn from unlabeled data. The ablation study further demonstrates our hypothesis (i.e., especially enforcing consensus in the hard ambiguous regions can maximize the efficacy of unlabeled data) and shows impressive results under various extreme labeled conditions.

2. Related work

2.1. Semi-supervised medical image segmentation

As summarized in Section 1, most semi-supervised segmentation methods feature two popular fashions, i.e., pseudo labeling (Zhang et al., 2022; Zheng et al., 2020; Bai et al., 2017; Rizve et al., 2021; Zhang et al., 2021a) and smoothness assumption based consistency regularization (Samuli and Timo, 2017; Tarvainen and Valpola, 2017; Cui et al., 2019; Luo et al., 2021; Wang et al., 2021; You et al., 2022; Li et al., 2020a; Ouali et al., 2020; Yu et al., 2019; Shi et al., 2021). Besides the above two dominant directions, there are also other perspectives to exploit the additional unlabeled data. For example, adversarial learning also shines in semi-supervised segmentation (Zhang et al., 2017; Sedai et al., 2017), wherein it encourages the predicted segmentation of unlabeled data to be closer to that of labeled data. Entropy minimization (Zhang et al., 2017) is adopted to enforce low-entropy predictions of the model to exploit the unlabeled data. Huang et al. (2021) used the bilateral graph convolution to capture the long-range dependencies and refine the semantic representation. CPCL (Xu et al., 2022c), which is based on the prototypical network in few-shot learning, utilizes non-parametric class-wise prototype learning to enable the real expert labels of labeled data to supervise unlabeled data.

Here, we focus on the category of consistency regularization. In this line, starting from the classical MT model (Tarvainen and Valpola, 2017), recent improved consistency-based methods are similarly dedicated to more complex data-level or model-level perturbations for unsupervised consistency learning (Li et al., 2020a; Ouali et al., 2020), incorporating self-supervised techniques (e.g., contrastive learning) as another regularization (You et al., 2022), advancing uncertainty estimation with conservative-radical networks (Shi et al., 2021) or introducing other parallel branches to provide multi-task guidance (Luo et al., 2021; Wang et al., 2021). These complex methods outperform the classical MT model by a considerable margin, leaving the impression that the classical MT model is too simple to learn productive representations from the unlabeled data. Yet, we find that the performance bottleneck is mainly due to indiscriminate (Cui et al., 2019) or low-yielding consistency learning (Yu et al., 2019). In this work, we put emphasis on further improving the classical MT model, where we rethink it from a neglected-yet-critical perspective, i.e., informative consistency target selection, and present the enhanced version to reach state-of-the-art performance.

2.2. Informativeness estimation

Although active learning (AL) requires the true labels (Budd et al., 2021), our work shares the similar spirit of selecting the most informative samples to be involved in the training process. AL aims to find out which samples are worthwhile to label rather than requiring experts to annotate all data (since many samples are redundant) while our work devotes to suggesting the worthwhile voxel-wise locations in the unlabeled data as the informative consistency targets for rewarding semi-supervised training. The most critical part in both AL and our work is how to effectively find out the ambiguous (but informative) samples (AL) or voxel-wise locations for consistency regularization (ours).

For AL, the leading family of informativeness estimation falls into uncertainty (Czolbe et al., 2021; Atzeni et al., 2022; Nguyen et al., 2022), arguing that we can gain more information from the sample with higher uncertainty (Budd et al., 2021; Settles and Craven, 2008). Inspired by this, we introduce predicted entropy and Monte-Carlo dropout-based model uncertainty as two of our target selection strategies. On the other hand, recent works (Czolbe et al., 2021; Chen et al., 2021a; Wu et al., 2021) revealed that model uncertainty in segmentation correlates with error-prone ambiguous regions. Therefore, we attempt to bridge pseudo labeling and consistency regularization, i.e., heuristically finding the errors in pseudo labels and imposing consistency learning on these locations. The recent work (Zhang et al., 2021b) on self-training based unsupervised domain adaptation demonstrated that class prototypes are less sensitive to the errors (assumed as the minority) in pseudo labels, and thus can help label denoising. Northcutt et al. (2021) proposed to ask for an additional model to help estimate the joint distribution matrix between given noisy labels and latent true labels to statistically characterize systematic label errors for the natural image classification task. Considering the substantial differences in tasks and goals between our work, here, we adapt the two spirits of label noise identification as plug-and-play modules into the MT model to discriminate the informative targets. Specifically, we regard the teacher as a self-ensembling third party to help find the

errors in student's pseudo labels. Note that informativeness estimation is still an open problem in both AL and our study.

3. Methodology

In this section, we present how to improve the classical MT model into AC-MT (Fig. 3), the enhanced version of MT that prioritizes informative target selection with state-of-the-art performance in semi-supervised medical image segmentation. Overall, AC-MT consists of the vanilla MT model (Section 3.1) and the ambiguity identification module (Section 3.3) which selects the high-yielding targets for the consistency loss.

3.1. Mean-teacher model

Regarding the semi-supervised segmentation task, we denote the labeled set as $\mathcal{D}_L = \left\{X_{l(i)}, Y_{l(i)}\right\}_{i=1}^{M}$ and the unlabeled set as $\mathcal{D}_U = \left\{X_{u(i)}\right\}_{i=M+1}^{N}$, where $X_{l(i)}, X_{u(i)} \in \mathbb{R}^{H\times W\times D}$ represent the scans with height H, width W, and depth D, and $Y_{l(i)} \in \{0,1\}^{H\times W\times D}$ denotes the label of $X_{l(i)}$ (we focus on binary task here). Following the vanilla MT framework (Cui et al., 2019), the objective of the semi-supervised segmentation task can be formulated as:

$$\min_{\theta} \mathcal{L}_s\left(\theta, \mathcal{D}_L\right) + \lambda \mathcal{L}_c(\theta, \tilde{\theta}, \mathcal{D}_U), \tag{1}$$

where $\mathcal{L}_s$ is the supervised loss for $\mathcal{D}_L$; $\mathcal{L}_c$ is the unsupervised consistency loss for $\mathcal{D}_U$; λ is a ramp-up weight commonly scheduled by the time-dependent Gaussian function $\lambda(t) = w_{max} \cdot e^{-5\left(1-\frac{t}{t_{max}}\right)^2}$ to avoid the domination by meaningless consistency targets at the early training stage, where w_{max} and $t_{\max}$ are the maximum consistency weight and training step, respectively; θ and $\tilde{\theta}$ are the weights of the student model and the teacher model, respectively. In the temporal-ensembling MT model, the teacher's weights $\tilde{\theta}_t$ at the training step t are updated by exponential moving average (EMA) of the student's weights θ_t, which can be formulated as:

$$\tilde{\theta}_t = \alpha\tilde{\theta}_{t-1} + (1-\alpha)\theta_t, \tag{2}$$

where α is the EMA decay rate and empirically set to 0.99 (Yu et al., 2019). When the same input is fed into both models, $\mathcal{L}_c$ encourages the teacher's prediction to be consistent with the student's under different perturbations ξ and ξ' (Luo et al., 2018). The most common and simplest practice, which we follow in this study, is the input-level perturbation, i.e., adding Gaussian noises to the input images (Yu et al., 2019). We follow (Yu et al., 2019; Xu et al., 2021; Cui et al., 2019) where $\mathcal{L}_c$ is measured by mean squared error (MSE) for fairness.

3.2. Seeking ambiguity consensus towards rewarding guidance

Distinct from the classical MT (Cui et al., 2019) that involves all voxels or the recent UA-MT (Yu et al., 2019) that selects reliable (yet less-informative) regions for the consistency loss calculation, we especially encourage consistency between the student's and the teacher's predictions at the identified ambiguous regions (see Section 3.3). Specifically, we re-design the consistency loss $\mathcal{L}_c$ as the ambiguity-masked mean squared error:

$$\mathcal{L}_c\left(f_t, f_s, E\right) = \frac{\sum_v \left[E_v \sum_c \left(\left\|f_{t,v,c} - f_{s,v,c}\right\|^2\right)\right]}{\sum_v E_v}, \tag{3}$$

where E_v is the binary ambiguity indicator at the vth voxel; $f_{t,v,c}$ and $f_{s,v,c}$ are the pre-softmax predictions of teacher model and student model for each class $c \in C$ at the vth voxel, respectively. With such ambiguity-aware consistency loss, the model can directly focus on learning the most informative clues from the unlabeled data by their consensus on these regions. Partly echoing the spirit in active learning (Budd et al., 2021) that finds the most informative samples to annotate, inherently, our method tries to find out the most worthwhile (challenging) regions to learn from their large volatility under perturbations (Fig. 2) and enables the model to learn more generalized representations for consistent predictions on them. Then, the next important question is how to effectively select the ambiguous-yet-informative consistency targets.

3.3. Strategies of informative target selection

The nature of ambiguity often arises from the characteristic of the voxel itself, where we aim to find out such ambiguity via different strategies. Regarding the design of the target selection strategies, two principles will be imposed: (i) the strategy should be model-agnostic, i.e., no change to the backbone, ensuring the plug-and-play flexibility for other relevant fields and the same representation potential of the network, and (ii) no extra trainable parameters should be introduced, e.g., no auxiliary trainable network, ensuring fair model complexity. Based on the two principles, we introduce a family of plug-and-play informative target selection strategies from the perspectives of entropy (Section 3.3.1), model uncertainty (Section 3.3.2) and label noise self-identification (Section 3.3.3 and Section 3.3.4) to suit various applications. A summarized discussion can be also found in Section 5.

3.3.1. High Softmax Entropy (Ent.)

The network backbone (e.g., V-Net (Milletari et al., 2016) and U-Net (Çiçek et al., 2016)) with a softmax output layer produces a voxel-wise class probability distribution. Denoting p_s as the predicted probability of the student model at a voxel, we can measure the normalized entropy ne_v for the voxel as:

$$ne_v = -\sum_{c\in C} p_s^c \log\left(p_s^c\right) / \log(|C|) \in [0,1], \tag{4}$$

where $c \in C$ is the semantic class. A high entropy value means that the model is disordered at this voxel, implying that this voxel is likely difficult. Thus, we select the voxels with $ne_v > H$ to obtain the entire ambiguity map $E \in \mathbb{R}^{H\times W\times D}$ for calculation of consistency loss at the current training step, where H is an empirical threshold. Formally, the ambiguity indicator E_v at the vth voxel is formulated as $E_v = \mathbb{1}\left[ne_v > H\right]$, where $\mathbb{1}(\cdot)$ is the indicator function. Considering that the model disorders could be reduced as the training goes, H ranges from 0.75 to 0.60 using a Gaussian ramp-down paradigm.

3.3.2. High model uncertainty (MU)

Model uncertainty is related to the model prediction for the input which may be inaccurate due to insufficient training/data, poor convergence, etc. High-uncertainty samples are often identified with various methods and then leveraged in active learning, which has been well demonstrated that high uncertainty usually correlates with high informativeness (Budd et al., 2021). Test-time Monte-Carlo dropout serves as an approximation of a Bayesian neural network (BNN) (Kendall and Gal, 2017), which interprets the probability distribution over the model's parameters. Technically, two dropout layers with dropout rate 0.5 are utilized at the last layer at Down-Stage and the first layer at Up-Stage, similar to Yu et al. (2019). Turning the dropout layers on, T stochastic forward passes are performed, which can be interpreted as Monte-Carlo samples of posterior distribution of the model's weights. Note that unlike UA-MT (Yu et al., 2019) using the weight-averaged teacher model for uncertainty measurement, we use the student model here to reflect the current model's uncertainty. By such, we obtain a set of probability vector $\left\{\mathbf{p}_s\right\}_{t=1}^{T}$, and we also adopt the normalized entropy as a measure of the uncertainty for each voxel:

$$u_v = -\sum_{c\in C}\left(\frac{1}{T}\sum_t \mathbf{p}_s^c\right)\log\left(\frac{1}{T}\sum_t \mathbf{p}_s^c\right) / \log(|C|) \in [0,1]. \tag{5}$$

Similarly, we select the high-uncertainty voxels with $u_v > H$ to obtain the entire ambiguity map $E \in \mathbb{R}^{H\times W\times D}$ for the image with the same range of H as in Section 3.3.1. Formally, the ambiguity indicator E_v at the vth voxel is formulated as $E_v = \mathbb{1}\left[u_v > H\right]$, where $\mathbb{1}(\cdot)$ is the indicator function.

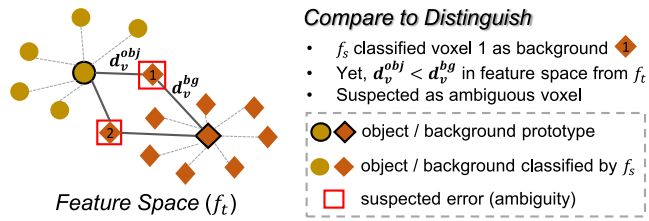

Fig. 4. Schematic diagram of the prototype-guided label noise identification.

3.3.3. Prototype-guided label noise identification (P-Err.)

Preliminaries. Label noise is the notorious obstacle in pseudo-labeling paradigm (Zhang et al., 2022; Zheng et al., 2020; Bai et al., 2017). Although our work does not fall into pseudo labeling, we notice that the errors of pseudo labels usually correlate with ambiguous regions where the current model encounters cognitive difficulties in the context of limited labeled data. Here, we bridge pseudo labeling and consistency regularization. Specifically, the student model further generates noisy pseudo labels on-the-fly using argmax operation and the teacher model serves as a self-ensembling party that enables label noise self-identification. We believe that these erroneous locations can promisingly discover the ambiguous-yet-informative voxel-wise locations for unidirectional consistency regularization. Note that no perturbation is employed in the label noise identification process as we aim to find the exact locations of informative voxels in the original images rather than the randomly perturbed images. Briefly, we attempt to obtain the ambiguity map by discriminating the student's errors assisted by the teacher model.

Based on our principles, firstly, we present a prototype-guided label noise self-identification strategy, abbreviated as "**P-Err.**". Inspired by self-training-based unsupervised domain adaptation (UDA) (Zhang et al., 2021b; Chen et al., 2021a) and self-learning from noisy labels (Han et al., 2019), this strategy leverages class prototypes, which are less sensitive to the outliers that are assumed to be the minority, to identify the errors in student's pseudo labels. Such a strategy is inherently motivated by the assumption that for a correct labeled voxel, its features should also lie closer to the corresponding class prototype, otherwise, a potential label noise is suspected. This spirit suits most medical images since the segmentation regions (e.g., organ and tumor) of the same class are often highly correlated. In addition, it has been demonstrated to be robust to real noises without requiring extra clean supervision or an accessorial network (Han et al., 2019; Chen et al., 2021a).

We illustrate a schematic diagram of the "P-Err". strategy in Fig. 4. Specifically, we adopt the confidence-weighted masked average pooling operation (Chen et al., 2021a; Xu et al., 2022c) to generate the class-wise prototype on-the-fly. Given an unlabeled image X_u and noisy pseudo label $\hat{Y}_s$ from the student model, we denote the feature map before the penultimate convolution in the teacher model as $F_t^p \in \mathbb{R}^{H_p \times W_p \times Z_p \times K}$ (K is the channel number), which is then upsampled to $F_t \in \mathbb{R}^{H \times W \times Z \times K}$ to be consistent with the size of segmentation mask via trilinear interpolation. The predicted one-hot labels $\hat{Y}_t$ of the self-ensembling teacher model (via the argmax operation) serve as the "mask" for the following prototype generation. Denoting P_t as the predicted probability of the teacher model, we compute the class prototypes q^{obj} and q^{bg} for object and background, respectively, via confidence-weighted masked average pooling as:

$$\begin{cases} q^{obj} = \dfrac{\sum_v \mathbb{1}[\hat{Y}_{t,v} \in C_{obj}] \cdot F_{t,v} \cdot P_{t,v}^{obj}}{\sum_v \mathbb{1}[\hat{Y}_{t,v} \in C_{obj}] \cdot P_{t,v}^{obj}}, \\ q^{bg} = \dfrac{\sum_v \mathbb{1}[\hat{Y}_{t,v} \in C_{bg}] \cdot F_{t,v} \cdot P_{t,v}^{bg}}{\sum_v \mathbb{1}[\hat{Y}_{t,v} \in C_{bg}] \cdot P_{t,v}^{bg}}, \end{cases} \tag{6}$$

where $\mathbb{1}(\cdot)$ is the indicator function. Then, the relative feature distances d_v^{obj} and d_v^{bg} between the vth feature vector $F_{t,v}$ and the prototypes q^{obj} and q^{bg} are calculated by:

$$d_v^{obj} = \left\| F_{t,v} - q^{obj} \right\|_2 \quad \text{and} \quad d_v^{bg} = \left\| F_{t,v} - q^{bg} \right\|_2 . \tag{7}$$

If the vth pseudo label $\hat{Y}_{s,v}$ from the student model is object (background) yet its feature vector $F_{t,v}$ lies closer to the background (object) prototype than the object (background) prototype, this voxel can be suspected as an erroneously labeled one. As such, the ambiguity map E is formulated as:

$$E_v = \mathbb{1}[\hat{Y}_{s,v} = 1] \cdot \mathbb{1}[d_v^{obj} \geq d_v^{bg}] + \mathbb{1}[\hat{Y}_{s,v} = 0] \cdot \mathbb{1}[d_v^{obj} \leq d_v^{bg}]. \tag{8}$$

3.3.4. Class-conditional systematic label noise identification (S-Err.)

In addition to P-Err., another alternative strategy of label noise self-identification resorts to the spirit in the recent work called confident learning (Northcutt et al., 2021) on class-conditional label noise identification, which revealed that the joint confident matrix, i.e., incorporating thresholds into the classical confusion matrix (Chen et al., 2019), is more robust to class-imbalanced and over-confident predictions, while its normalized statistics (i.e., joint distribution matrix between the observed and latent true labels) can effectively reflect the aleatoric uncertainty from latent label noises (detailed in Theorem 2 in Northcutt et al. (2021)) under the classification noise process (CNP) (Angluin and Laird, 1988) assumption. Based on this spirit, Northcutt et al. (2021) have successfully identified substantial incorrect ground truth in mainstream natural image datasets such as CIFAR (Krizhevsky et al., 2009) and ImageNet (Russakovsky et al., 2015). Note that such spirits were originally proposed for the image classification task. To suit our image segmentation task, we adapt the CNP assumption, that is, we assume that every voxel $x \in X$ has a true (latent) label y^*, while every label in class $j \in L$ can be *independently* mislabeled as class $i \in L$ with probability $p(y = i \mid y^* = j)$. This assumption is reasonable and has been used in prior work (Goldberger and Ben-Reuven, 2017; Sukhbaatar et al., 2015). Further, we can take the spirit of Random Walk (Von Luxburg, 2007) to estimate the joint probability $p(y = i, y^* = j)$ to compute the possible quantity of mislabeled voxels in the image X. Here, such spirit is elegantly tailored in the MT model to effectively discriminate the informative targets. Note that this strategy relies on an external prediction, as such, it specializes finding "systematic errors" (e.g., a fox is much more likely to be labeled as dog than cow, and here, if there are non-relevant tissues with similar color or texture characteristics as the left atrium in the image, they are more likely to be mislabeled as part of the left atrium by the student model trained with limited labeled data). Thus, for convenience, we abbreviate this strategy as "**S-Err.**". Here, considering the weight-averaged predictions P_t from the teacher model f_t has the ensembling nature, we take it to construct the desired matrices. As such, we derive the student's one-hot predictions $\hat{Y}_s$ as the observed noisy label. Given basic supervision by the labeled data, the label noises in $\hat{Y}_s$ are usually systematic. In short, f_t continuously provides the self-ensembling probabilities P_t to differentiate the errors in the one-hot predictions $\hat{Y}_s$ from f_s, so that the consistency targets can be divided by two groups.

An illustrative toy example of this strategy is shown in Fig. 5. Specifically, we denote the probability $p_t^i(x)$ that a voxel x belongs to the student's pseudo label $\hat{y}_s = i$ as its consensus confidence (e.g., yellow for class 0 ($P_t^0[\hat{Y}_s == 0]$) in Fig. 5). Intuitively, if x with $\hat{y}_s = i$ satisfies $p_t^j(x) \geq \gamma^j$, the latent true label y^* of x can be suspected to be j rather than i. Given $\hat{Y}_s$ and suggested by Northcutt et al. (2021), we use the mean confidence as the per-class threshold for each image X, i.e., $\gamma^j = \text{mean}(P_t^j[\hat{Y}_s == j])$. With this threshold, we estimate the joint confident matrix $\mathbf{C}_{\hat{y}_s, y^*}$, a matrix obtained after counting the number of voxels that meet the per-class threshold, for the image X as:

$$\mathbf{C}_{\hat{y}_s, y^*}[i][j] = \left| \hat{X}_{\hat{y}_s, y^{j*}} \right| = \left| \left\{ x \in X_{\hat{y}_s = i} : p_t^j(x) \geq \gamma^j, j = \underset{l \in L : p_t^l(x) \geq \gamma^l}{\arg\max} \; p_t^l(x) \right\} \right|, \tag{9}$$

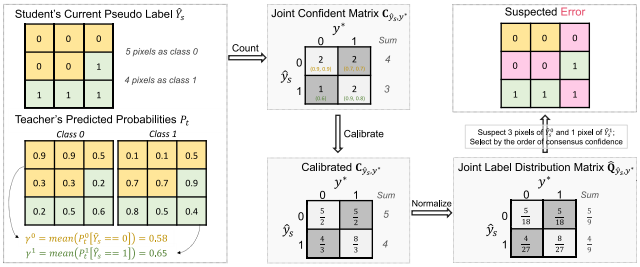

Fig. 5. An illustrative toy example of the S-Err. strategy.

where L denotes the semantic classes. We calibrate $\mathbf{C}_{\hat{y}_s,y^*}$ since the threshold filters out many voxels, and then normalize the calibrated $\mathbf{C}_{\hat{y}_s,y^*}$ to obtain the $L \times L$ joint label distribution matrix $\hat{\mathbf{Q}}_{\hat{y}_s,y^*}[i][j]$:

$$\hat{\mathbf{Q}}_{\hat{y}_s,y^*}[i][j] = \frac{\frac{\mathbf{C}_{\hat{y}_s,y^*}[i][j]}{\sum_{j\in L}\mathbf{C}_{\hat{y}_s,y^*}[i][j]} \cdot \left|X_{\hat{y}_s=i}\right|}{\sum_{i\in L,j\in L}\left(\frac{\mathbf{C}_{\hat{y}_s,y^*}[i][j]}{\sum_{j\in L}\mathbf{C}_{\hat{y}_s,y^*}[i][j]} \cdot \left|X_{\hat{y}_s=i}\right|\right)}, \tag{10}$$

where $\left|X_{\hat{y}_s=i}\right|$ is the number of voxels with label $\hat{y}_s = i$. Note that $\mathbf{C}_{\hat{y}_s,y^*}$ and $\hat{\mathbf{Q}}_{\hat{y}_s,y^*}$ can be estimated for the entire image batch during training since this estimation is closer to the real distribution as the size of the data increases. Then, taking the spirit of Random Walk (Von Luxburg, 2007), the off-diagonal elements of $\hat{\mathbf{Q}}_{\hat{y}_s,y^*}$ indicate how likely the classes are mislabeled, and low consensus confidence can be a heuristic likelihood of being a mislabeled voxel. Therefore, $\texttt{round}\left(|X| \cdot \sum_{j\in L:j\neq i}\left(\hat{\mathbf{Q}}_{\hat{y}_s,y^*}[i][j]\right)\right)$ voxels with the lowest consensus confidence for each class i are selected as the mislabeled ones, where $|X|$ denotes the number of voxels in X. As marked in pink in Fig. 5, three pixels labeled as the background are suspected, while one pixel labeled as the object is also suspected. In this way, we obtain the voxel-level error map $E \in \{0,1\}^{H\times W\times D}$, where "1" represents that $\hat{y}_s$ is suspected as an error at this training step, so that the student's predictions can be divided into the error group and the correct group. The error group, as exampled in Fig. 3, can be regarded as the ambiguous voxels.

4. Experiments

4.1. Datasets and experimental setup

4.1.1. Left atrium segmentation dataset

First, following the mostly used benchmark in semi-supervised segmentation (Yu et al., 2019; Luo et al., 2021; Wang et al., 2020b), we perform extensive evaluation on the left atrium (LA) segmentation dataset (Xiong et al., 2021) which contains 100 3D gadolinium-enhanced magnetic resonance images (GE-MRIs). The images have the isotropic resolution of $0.625 \times 0.625 \times 0.625$ mm^3. Fairly, we follow the same data division protocol and preprocessing procedure as in Yu et al. (2019), Luo et al. (2021), Wang et al. (2020b), Xu et al. (2022a), where 80 samples are used for training and the remaining 20 samples are used for testing; the images are cropped to the center of heart region and normalized to zero mean and unit variance.

4.1.2. Brain tumor segmentation dataset

Besides the most commonly used LA benchmark, we also evaluate the methods on whole brain tumor segmentation from T2-FLAIR MRI, following (Xu et al., 2022c). This dataset is from the BraTS 2019 challenge (Spyridon (Spyros) Bakas, 2020), containing preoperative MRI (with modalities of T1, T1Gd, T2 and T2-FLAIR) of 335 glioma patients, where 259 patients are with high-grade glioma (HGG) and 76 patients are with low-grade glioma (LGG). Being critical to brain tumor surgery, T2-FLAIR can well characterize the malignant tumors (Zeineldin et al., 2020), and thus we use T2-FLAIR images in this task. The images are resampled to the isotropic resolution of $1 \times 1 \times 1$ mm^3. Following the same data split and preprocessing procedure as in Xu et al. (2022c), 250 samples are used for training, 25 for validation and the remaining 60 for testing.

4.1.3. Baseline approaches

The supervised-only (SupOnly) models serve as the baselines. Besides the most relevant SSL approaches, i.e., the original mean-teacher model (Cui et al., 2019) and uncertainty-aware MT (UA-MT) (Yu et al., 2019), we also include diverse state-of-the-art semi-supervised segmentation methods, including deep adversarial network (DAN) (Zhang et al., 2017), cross pseudo supervision (CPS) (Chen et al., 2021b), entropy minimization (Entropy Mini) (Vu et al., 2019), double-uncertainty weighting (DUW) (Wang et al., 2020b), interpolation consistency training (ICT) (Verma et al., 2022), shape-aware model (SASSNet) (Li et al., 2020b), dual-task consistency (DTC) (Luo et al., 2021), cross-consistency training (CCT) (Ouali et al., 2020), uncertainty rectified pyramid consistency (URPC) (Luo et al., 2022) and cyclic prototypical consistency learning (CPCL) (Xu et al., 2022c). All the methods are implemented with the same backbone and training protocols to ensure fairness. The baseline approaches have been implemented in our comprehensive SSL library: https://github.com/HiLab-git/SSL4MIS.

4.1.4. Implementation details and evaluation metrics

The framework is based on the PyTorch implementation of Yu et al. (2019) using an NVIDIA GeForce RTX 3090 GPU. The supervised loss uses the joint cross-entropy loss and Dice loss (Yu et al., 2019). The batch size is set to 4 including 2 labeled volumes and 2 unlabeled volumes. The maximum consistency weight w_{max} is empirically set to 0.1 (Yu et al., 2019). The network is trained using the SGD optimizer (weight decay = 0.0001, momentum = 0.9). The learning rate is initialized as 0.01 and decayed with a power of 0.9 after each iteration. Data augmentation, including randomly flip and rotation, is applied. For LA segmentation, we adopt the same 3D V-Net backbone (Milletari et al., 2016) as in Yu et al. (2019), Li et al. (2020b), Luo et al. (2021). Consistently, we randomly crop patches of $112 \times 112 \times 80$ voxels as the input and use sliding window strategy with stride of $18 \times 18 \times 4$ voxels for prediction. $t_{\max}$ is set to 6,000. For brain tumor segmentation, we follow (Xu et al., 2022c) that uses the 3D U-Net backbone (Çiçek et al., 2016) for a fair comparison. We randomly crop patches of $96 \times 96 \times 96$ voxels as the input and use sliding window strategy with stride of $64 \times 64 \times 64$ voxels for prediction. $t_{\max}$ is set to 20,000. The model is initialized using Kaiming initialization (He et al., 2015). The teacher model is updated by exponential moving average (EMA) of the student's weights.

For a fair comparison, no extra post-processing or ensemble methods are utilized. We adopt four metrics for a comprehensive evaluation, including Dice score, Jaccard, average surface distance (ASD) and 95% Hausdorff distance (95HD). The implementation will be available at https://github.com/lemoshu/AC-MT.

4.2. Experiments on left atrium segmentation

4.2.1. Comparison with state-of-the-art methods

Table 1 presents the performance of our methods and other approaches on LA segmentation under the commonly used 10% and 20% labeled data settings (Yu et al., 2019; Wang et al., 2020b; Luo et al., 2021). It can be observed that our four target selection strategies consistently achieve competitive results compared with other lines of approaches, and bring substantial improvements over the original MT (Tarvainen and Valpola, 2017) and UA-MT (Yu et al., 2019) under both 10% and 20% labeled data settings without introducing extra tasks or more trainable parameters. Especially, the fourth strategy, i.e., AC-MT (S-Err.), achieves the best performance under both settings, indicating that this strategy can better distinguish the informative

Table 1
Quantitative comparison on the LA dataset (Xiong et al., 2021). † indicates our best-performing strategy in this task. ∗ and ∗∗ indicate $p \leq 0.05$ and $p \leq 0.01$ from a two-sided paired t-test when comparing our best model with others, respectively. The best results are shown in bold.

Method	# Scans Used		Metrics			
	Labeled	Unlabeled	Dice [%] ↑	Jaccard [%] ↑	ASD [voxel] ↓	95HD [voxel] ↓
SupOnly	10%	0	79.99**	68.12**	5.48**	21.11**
DAN (Zhang et al., 2017)	10%	90%	75.11**	63.47**	3.57**	19.04**
CPS (Chen et al., 2021b)	10%	90%	84.09**	73.17**	2.41	22.55**
Entropy Mini (Vu et al., 2019)	10%	90%	85.90**	75.60**	2.74*	18.65**
DUW (Wang et al., 2020b)	10%	90%	85.91**	75.75**	3.31**	12.67*
ICT (Verma et al., 2022)	10%	90%	85.39**	74.84**	2.88*	17.45**
SASSNet (Li et al., 2020b)	10%	90%	86.81*	76.92**	3.94**	12.54*
DTC (Luo et al., 2021)	10%	90%	86.57**	76.55**	3.74**	14.47**
CCT (Ouali et al., 2020)	10%	90%	85.49**	76.45**	2.99*	15.36**
URPC (Luo et al., 2022)	10%	90%	85.02**	75.98**	2.95*	15.21**
CPCL (Xu et al., 2022c)	10%	90%	86.08**	76.89**	2.33	12.37*
MT (Tarvainen and Valpola, 2017)	10%	90%	84.24**	73.26**	2.71*	19.41**
UA-MT (Yu et al., 2019)	10%	90%	84.25**	73.48**	3.36**	13.84*
Ours: AC-MT (Ent.)	10%	90%	87.52	77.93	2.91	10.87
Ours: AC-MT (MU)	10%	90%	86.74	76.81	2.57	14.73
Ours: AC-MT (P-Err.)	10%	90%	87.33	77.68	2.52	14.38
Ours: AC-MT (S-Err.)†	10%	90%	**89.12**	**80.46**	**2.19**	**11.05**
SupOnly	20%	0	86.03**	76.06**	3.51**	14.26**
DAN (Zhang et al., 2017)	20%	80%	87.52**	78.29**	2.42**	9.01*
CPS (Chen et al., 2021b)	20%	80%	87.87**	78.61**	2.16*	12.87**
Entropy Mini (Vu et al., 2019)	20%	80%	88.45*	79.51**	3.90**	14.14**
DUW (Wang et al., 2020b)	20%	80%	89.65	81.35*	2.03	7.04*
ICT (Verma et al., 2022)	20%	80%	89.02*	80.34**	1.97	10.38**
SASSNet (Li et al., 2020b)	20%	80%	89.27*	80.82**	3.13**	8.83**
DTC (Luo et al., 2021)	20%	80%	89.42	80.98**	2.10*	7.32*
CCT (Ouali et al., 2020)	20%	80%	88.01**	80.95**	2.37**	8.25**
URPC (Luo et al., 2022)	20%	80%	88.43*	81.15*	2.35*	8.21**
CPCL (Xu et al., 2022c)	20%	80%	88.32*	81.02*	2.02	8.01*
MT (Tarvainen and Valpola, 2017)	20%	80%	88.42*	79.45*	2.73**	13.07**
UA-MT (Yu et al., 2019)	20%	80%	88.88*	80.21*	2.26*	7.32*
Ours: AC-MT (Ent.)	20%	80%	89.63	81.32	1.89	10.17
Ours: AC-MT (MU)	20%	80%	89.64	81.34	1.94	8.29
Ours: AC-MT (P-Err.)	20%	80%	89.58	81.23	1.93	7.51
Ours: AC-MT (S-Err.)†	20%	80%	**90.31**	**82.43**	**1.76**	**6.21**
SupOnly (upper bound)	100%	0	91.14	83.82	1.52	5.75

regions in this task (will be elaborated in Section 4.2.3(i)). Specifically, AC-MT (S-Err.) brings around 5.8% and 9.5% improvements over UA-MT (Yu et al., 2019) in terms of Dice and Jaccard metrics under the challenging 10% labeled setting, respectively. It also outperforms other state-of-the-art methods by a large margin under the 10% setting, further demonstrating that the target selection strategy matters in the MT model. By effectively discriminating more informative clues from the unlabeled data for consistency regularization, more potentials of the MT model can be exploited. Fig. 6 presents exemplar results of our best-performing model variant in this task, i.e., AC-MT (S-Err.), and the two most relevant approaches under the 10% labeled data setting. Consistently, the predictions of our AC-MT (S-Err.) fit more accurately with the ground-truth masks, further verifying the effectiveness of our strategy.

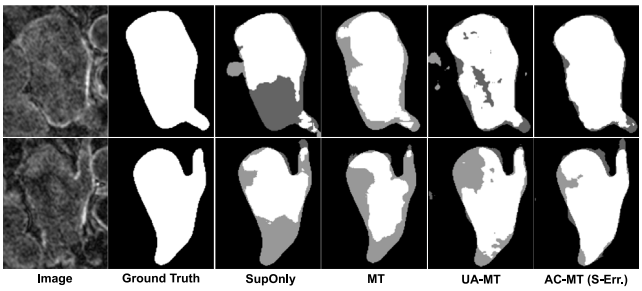

Fig. 6. LA segmentation results (shown on exemplar 2D slices) with only 10% labeled data for training. Gray color represents the inconsistency between the predicted result and the ground truth.

4.2.2. Extremely scarce labeling budgets

Previous works (Luo et al., 2021; Yu et al., 2019; Wang et al., 2020b) on the LA segmentation benchmark only focused on the aforementioned 10% and 20% labeled data settings, whereas we are also interested in some extreme situations. Table 2 presents the results under the most extreme setting, i.e., only 2 labeled samples (the minimum labeled batch size) are available. It can be observed that our methods still perform admirably under the most extreme condition. Especially, our best variant in this task, i.e., AC-MT (S-Err.), yields a Dice score of 81.18%. In contrast, MT and UA-MT only achieve 72.78% and 74.63%, respectively. Fig. 7 further presents the performance under varying labeling budgets as measured by Dice score, wherein our four strategies consistently improve the MT model. Such experiments further demonstrate the robustness of our strategies and indicate that when confronted with extremely scarce labeling budgets, emphatically enforcing invariant predictions in ambiguous regions can be more beneficial. Especially, AC-MT (S-Err.) and AC-MT (P-Err.) achieve better results than AC-MT (Ent.) and AC-MT (MU) when less than 10% labeled data are available, revealing that finding informative errors from student's pseudo labels is more appealing when very limited labeling budgets are accessible.

4.2.3. Ambiguity Consensus (AC) vs. Determinacy Consensus (DC)

Effectively discriminating the informative ambiguous group is the key to the effectiveness of ambiguity-consensus learning. Qualitatively,

Table 2
Comparison under the most extreme setting (2.5% (2 samples) are labeled scans and 97.5% are unlabeled scans). The best results are in bold.

Method	Metrics			
	Dice [%] ↑	Jaccard [%] ↑	ASD [voxel] ↓	95HD [voxel] ↓
SupOnly	70.46	56.32	3.74	31.13
MT (Cui et al., 2019)	72.78	58.88	3.65	34.16
UA-MT (Yu et al., 2019)	74.63	60.79	3.39	35.98
AC-MT (Ent.)	78.25	64.93	3.23	30.25
AC-MT (MU)	75.98	62.61	3.87	31.17
AC-MT (P-Err.)	81.09	68.67	**2.88**	**28.14**
AC-MT (S-Err.)	**81.18**	**68.72**	3.18	29.16

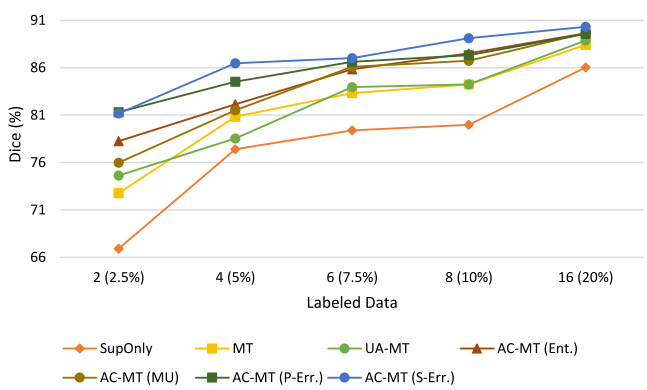

Fig. 7. Dice scores on LA segmentation under varying labeling budgets.

the examples of estimated ambiguity map E from four strategies at training iteration 1,000 with 10% labeled data are visualized in Fig. 8. In Fig. 9, we also provide an additional example of the logit difference image and the ambiguity map (overlapping on the logit difference image with 80% opacity) throughout the training process, with a 3D rendering of ambiguity map in the top right. As observed, the ambiguous regions are generally the minority and often exist in some error-prone regions, e.g., small branches or blurred edges of the organ, and the two strategies based on label noise identification, particularly S-Err, identify more detailed regions to focus on. Note that the ambiguity map is obtained on the original image using different strategies, and the regions with large perturbed volatility do not necessarily represent the most informative ones (such as in some non-selected easy background regions that are not region-of-interest in Fig. 9). To quantitatively analyze their efficacy, DC-MT (determinacy-consensus mean-teacher model) is further introduced as the reverse version of AC-MT. Table 3 compares the results of our AC-MT and the reverse version DC-MT under the 10% labeled setting, where most metrics of AC-MT are significantly better than DC-MT ($p \leq 0.05$ from a two-sided paired t-test), which further demonstrates the effectiveness of our hypothesis that the error-prone ambiguous regions in unlabeled data contain more informative and valuable clues, while emphatically encouraging prediction consensus on these challenging regions can help the network capture more generalized features for rewarding training. Especially, the largest performance gap can be observed between AC-MT (S-Err.) and DC-MT (S-Err.), indicating that this strategy can more distinctively distinguish informative voxels from the pool.

4.3. Experiments on brain tumor segmentation

We notice that brain tumor segmentation is much more difficult than LA segmentation due to intractable tumor heterogeneity. Table 4 presents the results of our method and other methods under 10% and 20% labeled settings. As observed, all semi-supervised methods bring improvements over the supervised-only baselines, indicating that the unlabeled data contains rich and diverse information. Our four target

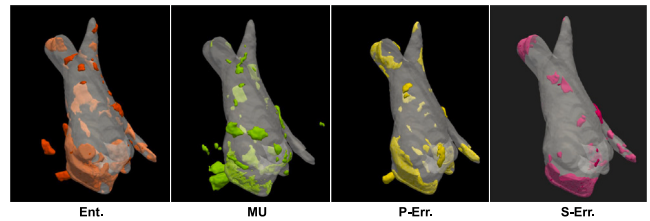

Fig. 8. 3D exemplar ambiguity maps E at training iteration 1,000 with 10% labeled data on LA segmentation. Gray color denotes ground truth.

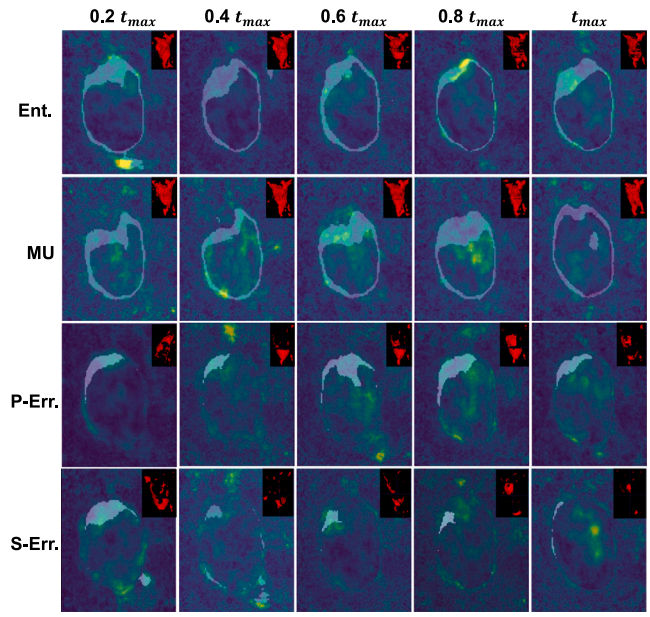

Fig. 9. 2D visualization of logit difference map and ambiguity map (overlapping with 80% opacity) during training, with a 3D rendering of ambiguity map in the top right.

Table 3
Comparison between ambiguity consensus and determinacy consensus for each target selection strategy on LA segmentation with 10% labeled data. $*$ indicates $p \leq 0.05$ from a two-sided paired t-test when comparing AC-MT with DC-MT for each strategy. The best results are in bold.

Method	Metrics			
	Dice [%] ↑	Jaccard [%] ↑	ASD [voxel] ↓	95HD [voxel] ↓
DC-MT (Ent.)	86.44	76.32	3.02	17.93
AC-MT (Ent.)	**87.52***	**77.93***	**2.91**	**10.87***
DC-MT (MU)	85.99	75.33	2.61	16.70
AC-MT (MU)	**86.74**	**76.81***	**2.57**	**14.73***
DC-MT (P-Err.)	86.23	76.48	2.66	18.10
AC-MT (P-Err.)	**87.33***	**77.68***	**2.52**	**14.38***
DC-MT (S-Err.)	86.28	76.14	2.88	14.77
AC-MT (S-Err.)	**89.12***	**80.46***	**2.19***	**11.05***

selection strategies still achieve competitive results compared with other approaches and bring substantial improvements over the original MT (Tarvainen and Valpola, 2017) and UA-MT (Yu et al., 2019) under the 10% labeled setting. We also notice that UA-MT (Yu et al., 2019) even performs worse than MT under the 10% labeled setting, further revealing that selecting certain voxels for consistency regularization can be suboptimal since most certain easy regions have been well learned via the guidance from limited labeled data. Overall, AC-MT (S-Err.) still achieves the best performance, further showing the effectiveness of this selection strategy. Specifically, under the 10% labeled setting, AC-MT (S-Err.) brings around 3.78% and 5.21% improvements over UA-MT (Yu et al., 2019) in terms of Dice and Jaccard metrics,

Table 4
Quantitative results on brain tumor segmentation (Spyridon (Spyros) Bakas, 2020). † indicates our best-performing strategy in this task. ∗ and ∗∗ indicate $p \leq 0.05$ and $p \leq 0.01$ from a two-sided paired t-test when comparing our best model with others, respectively. The best results are bold.

Method	# Scans Used		Metrics			
	Labeled	Unlabeled	Dice [%] ↑	Jaccard [%] ↑	ASD [voxel] ↓	95HD [voxel] ↓
SupOnly	10%	0	74.43**	61.86**	2.79**	37.11**
(Zhang et al., 2017)	10%	90%	81.71**	71.43*	2.32*	15.15**
CPS (Chen et al., 2021b)	10%	90%	82.52*	72.66*	2.66*	13.08*
Entropy Mini (Vu et al., 2019)	10%	90%	82.27*	72.15*	2.42*	11.98
ICT (Verma et al., 2022)	10%	90%	77.60**	67.72**	3.76**	15.19**
DTC (Luo et al., 2021)	10%	90%	81.75*	71.63*	2.56*	15.73**
CCT (Ouali et al., 2020)	10%	90%	82.15*	71.59*	2.11*	16.57**
URPC (Luo et al., 2022)	10%	90%	82.59*	72.11*	3.72**	13.88*
CPCL (Xu et al., 2022c)	10%	90%	83.36	73.23	1.99	11.74
MT (Tarvainen and Valpola, 2017)	10%	90%	81.94*	71.67*	2.33*	13.62*
UA-MT (Yu et al., 2019)	10%	90%	80.72**	70.30**	2.72**	11.76
Ours: AC-MT (Ent.)	10%	90%	83.03	72.97	2.01	12.56
Ours: AC-MT (MU)	10%	90%	83.18	73.24	1.97	12.43
Ours: AC-MT (P-Err.)	10%	90%	82.76	72.34	2.23	13.87
Ours: AC-MT (S-Err.)†	10%	90%	**83.77**	**73.96**	**1.93**	**11.37**
SupOnly	20%	0	80.16**	71.55**	3.43**	22.68**
DAN (Zhang et al., 2017)	20%	80%	83.31*	73.53**	2.23	10.86*
CPS (Chen et al., 2021b)	20%	80%	84.01*	74.02*	2.18	12.16*
Entropy Mini (Vu et al., 2019)	20%	80%	82.91*	73.15**	2.48	10.52*
ICT (Verma et al., 2022)	20%	80%	81.76**	72.01**	2.82*	9.66
DTC (Luo et al., 2021)	20%	80%	83.43*	73.56**	2.34*	14.77*
CCT (Ouali et al., 2020)	20%	80%	82.53**	72.36**	2.21	15.87**
URPC (Luo et al., 2022)	20%	80%	82.93*	72.57**	4.19**	15.93**
CPCL (Xu et al., 2022c)	20%	80%	83.48*	74.08	2.08	9.53
MT (Tarvainen and Valpola, 2017)	20%	80%	82.96*	72.95**	2.32*	9.85
UA-MT (Yu et al., 2019)	20%	80%	83.12*	73.01**	2.30	9.87
Ours: AC-MT (Ent.)	20%	80%	83.57	73.81	2.13	10.11
Ours: AC-MT (MU)	20%	80%	84.20	74.30	2.03	9.52
Ours: AC-MT (P-Err.)	20%	80%	84.16	74.34	**1.99**	9.87
Ours: AC-MT (S-Err.)†	20%	80%	**84.63**	**74.39**	2.11	**9.50**
SupOnly (upper bound)	100%	0	85.93	76.81	1.93	9.85

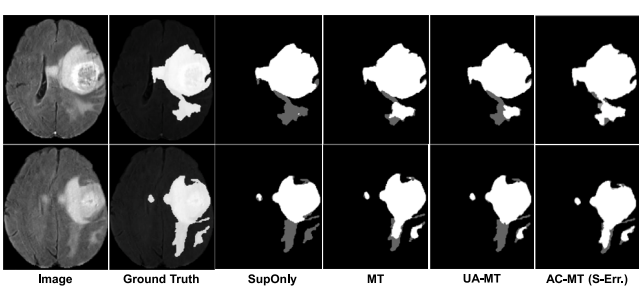

Fig. 10. Brain tumor segmentation results (shown on exemplar 2D slices) with only 10% labeled data. Gray color represents the inconsistency between the predicted result and the ground truth.

respectively. It also outperforms other state-of-the-art approaches with most metrics being significantly different ($p < 0.05$), demonstrating that especially enforcing invariant predictions on those ambiguous regions matters in semi-supervised learning. Fig. 10 presents exemplar results of our AC-MT (S-Err.) and the two most relevant approaches under the 10% labeled setting. As observed, the segmentation of our AC-MT (S-Err.) fit more accurately with the ground-truth masks, further verifying the effectiveness of our strategy.

5. Discussions

5.1. Choice of target selection strategy

Considering that each task has its inherent challenges and different computation resources in practice, we remold four plug-and-play strategies for ambiguous target selection from perspectives of entropy, model uncertainty and label noise self-identification, respectively, without introducing any trainable parameters or modifying the backbones. As observed in Table 2 and Fig. 7, AC-MT (S-Err.) and AC-MT (P-Err.) perform better than AC-MT (Ent.) and AC-MT (MU) under very limited labeling budgets, showing that label noise self-identification is an appealing way for informativeness estimation. This observation can also motivate future research in active learning. As observed in Table 1 and Table 4, AC-MT (S-Err.) consistently achieves the best performance, indicating that this strategy can better find the most informative regions. Table 3 further verifies that AC-MT (S-Err.) can more effectively categorize the voxels into two distinct groups, i.e., the redundant easy group and the informative hard group. Despite the superior performance, a common limitation of the two label noise self-identification strategies is that they require more computation resources and training time due to on-the-fly class prototype calculation (P-Err.) or joint confident matrix construction (S-Err.). Training AC-MT (S-Err.) and AC-MT (P-Err.) are experimentally ∼1.6x and ∼1.5x slower than that of AC-MT (Ent.) but it is still acceptable in most scenarios. On the other hand, AC-MT (Ent.) and AC-MT (MU) are computation-efficient, especially AC-MT (Ent.). They even slightly outperform AC-MT (P-Err.) when the labeled data increases to 10% and more. In addition, as shown in Fig. 7, AC-MT (Ent.) performs better than AC-MT (MU) when scarce labeling budgets are available. Such observations also exist in our further experiments on brain tumor segmentation. One possible reason is that the model cannot provide discriminative embedding space with extremely limited label supervision thus the predictions are full of randomness. Monte-Carlo dropout further exacerbates such randomness and thus it resembles a random sampling strategy. As the number of labeled samples increases, their performances become closer. Overall, we recommend AC-MT (S-Err.) for its robustness and superior performance. If training time matters and decent labeling budgets are available, AC-MT (Ent.) and AC-MT (MU) are also productive choices.

5.2. Impacts, future works and limitations

The MT model along with consistency regularization has been widely used in other applications, e.g., weakly-supervised learning (Gao et al., 2022) and learning with noisy labels (Wang et al., 2020a). This study can facilitate many relevant fields which leverage consistency regularization as a part of the learning objective. Besides, how to effectively and efficiently find the most informative targets is still an open problem in our study, also in the field of active learning. Our method can also shine to the active learning community that label noise self-identification can be a promising way to select high-informative samples to annotate. More collaboration among active learning, label noise identification in the field of noisy label learning and consistency-based semi-supervised learning is a promising direction. Regarding the limitations of our method, firstly, we speculate that our approach may not be directly applicable to the recent holistic scheme that combine pseudo labeling and consistency regularization, which is popularized by FixMatch (Sohn et al., 2020; Liu et al., 2022) for natural images. FixMatch employs the mean-teacher framework along with pseudo labeling, where one-hot pseudo labels from weakly perturbed unlabeled images (e.g., Gaussian noise, random rotations and translations) are used to supervise the prediction of strongly perturbed unlabeled images with noticeable modifications (e.g., color jitter, CutMix and context augmentation). The success of this approach can be attributed to that the model is more inclined to generate accurate predictions on weakly perturbed images. Our approach may not be directly applicable to FixMatch-based framework but UA-MT may work well (Yu et al., 2019), as it finally relies on the pseudo labeling scheme, which can be regard as *directional* consistency learning. However, it is important to note that the common strong perturbation techniques used in natural images may not be easily applicable to medical images. As such, the regular mean-teacher framework with the naïve unidirectional consistency is still the common and effective scheme in medical imaging community. Secondly, our strategy relies heavily on the teacher and student models themselves to identify informative ambiguous voxels. Another potential failure case is that if the model's discriminability is too weak for a specific task, the identified informative locations may be meaningless and may even exacerbate the model's biases. In this case, we may need to enhance the model's basic discriminability, for example by providing more labeled data for supervised training.

6. Conclusion

In this work, we identified a neglected-yet-critical problem, i.e., consistency target selection, that prevents the classical mean-teacher-based semi-supervised medical image segmentation framework from reaching its maximum potential, and then improved the mean-teacher model to a novel ambiguity-consensus mean-teacher (AC-MT) model along with four ambiguity identification strategies. Our key insight is to find out the most worthwhile pixel-wise consistency targets to learn from the unlabeled data and enable the model to especially learn more generalized representations on these challenging but informative regions. The superior performance on the left atrium segmentation and brain tumor segmentation demonstrated the effectiveness of our approach.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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